# AUTOMATED DETECTION AND CLASSIFICATION OF RICE LEAF DISEASES USING DEEP LEARNING

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Presented By

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**July, 2025**

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**July, 2025**

# Declaration

I, Md. Abdul Alim, hereby attest that the research described in this thesis, "Automated Detection and Classification of Rice Leaf Diseases Using Deep Learning," is the outcome of my own original work, which I conducted under the guidance of Most. Rehana Khatun. No other university or institution has received this thesis, in whole or in part, for the purpose of awarding a degree or diploma. All information and support sources used in this study have been properly credited. I attest that this work complies with academic integrity standards and is free of plagiarism.

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Acknowledgement

I would like to express my sincere gratitude to my supervisor Mst. Rehana Khatun, Asst. Professor, Department of Computer Science & Engineering, for her unwavering support, knowledgeable advice, and insightful recommendations during the writing of this thesis. Their support and enlightening criticism have been crucial in determining the course and results of this study. I genuinely value the commitment and support of the Computer Science & Engineering (CSE) faculty and staff in providing the necessary resources and fostering a conducive academic environment. I want to thank my friends and classmates for their cooperation and encouragement. I am most grateful to my parents and family, whose unwavering love, support, and prayers have been my biggest source of strength during this journey.

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# ABSTRACT

Rice is among the world's most important staple crops, particularly in Asia. But its outturn is heavily impacted by numerous leaf diseases that reduce crop quality and yield. Timely identification and precise categorization of these sickness can help farmers take timely actions. In this thesis, we propose a deep learning-based automatic rice leaf disease identification and categorization system. We used a CNN-based approach with experiments on models VGG19, and a custom Sequential CNN model. Our dataset used thousands of labeled images from publicly available rice disease datasets achieved an accuracy of 97% (Sequential CNN) and an accuracy of 91% (VGG19). The system has promising performance and can be incorporated into mobile applications to facilitate real-time disease diagnosis in the field. We employed image preprocessing methods like resizing, normalization, and augmentation to utilized the model performance. We tested the model's performance using typical classification metrics and confusion matrices. Based on our findings, this system can assist in reducing the need for expert consultation and manual inspection. Overall, the suggested method enhances the expanding field of AI-driven smart agriculture and offers a scalable solution for rice cultivation disease management.

**Keywords:** CNN, Deep Learning, Image Processing, Feature Extraction, Rice Disease Detection, Classification, Real Time Detection, Rice Leaf Analysis, Automated Diagnosis

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# Chapter 1 : Introduction

## 1.1 Introduction

Rice is a vital staple food and agricultural commodity in countries like Bangladesh. Its yield, however, is being threatened by leaf diseases such as Bacterial Leaf Blight, Brown Spot, Leaf Blast, Leaf Scald, Narrow Brown Spot, Neck Blast, Rice Hispa, Sheath Blight and Tungro. In the last 11 years, the part of agriculture in Bangladesh's Gross Domestic Product (GDP) was contributing about 15.08 percent. But unfortunately, the farmers who are working so hard to grow this crop, have to face huge losses because of crop damages caused by various diseases of paddy. [1]

Fig : Bacterial Leaf Blight

Fig : Brown Spot

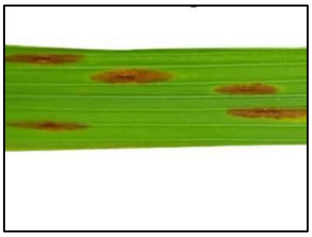


Fig : Narrow Brown Spot

Fig : Leaf Blast

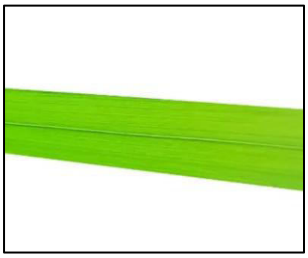


Fig : Healthy

Fig : Neck Blast

**Bacterial Leaf Blight :**

Bacterial Leaf Blight is a serious rice disease caused by Xanthomonas oryzae pv. oryzae. It usually shows up as yellowish, water-soaked spots that later become brown and dry. The disease spreads quickly in warm, humid weather, particularly through rain and irrigation water. If it is not controlled early, it can cause major yield loss.

**Brown Spot :**

Brown Spot is a fungal disease caused by Bipolaris oryzae. It is common in fields that lack nutrients or are under drought stress. Symptoms include round to oval brown spots with a yellow halo on the leaves. This disease weakens the plant and reduces both grain size and quality. It thrives in poor soil and low potassium conditions.

**Narrow Brown Spot :**

Narrow Brown Spot, caused by Cercospora oryzae, creates narrow, linear brown lesions on leaves and sheaths. It usually occurs during the late growth stages with prolonged high humidity. While it is less aggressive than other diseases, it can still impact photosynthesis and plant health. Constant wet conditions encourage its spread in rice fields.

Fig : Sheath Blight

Fig : Leaf Scald

Fig : Rice Hispa

Fig : Tungro

**Leaf Blast :**

Leaf Blast is a serious fungal disease caused by Magnaporthe oryzae. It shows up as diamond-shaped grayish spots with dark edges on rice leaves. The disease spreads quickly in warm, wet weather and with high nitrogen use. Severe infections can kill the plant and lead to significant yield losses.

**Leaf Scald :**

Leaf Scald is caused by the fungus Microdochium oryzae, which affects the leaves and sheaths. It appears as large, water-soaked lesions that turn pale brown with uneven edges. The disease usually develops in long wet conditions and impacts mature plants. It weakens the leaves, which reduces their ability to photosynthesize.

**Neck Blast :**

Neck Blast is a key phase of blast disease that affects the panicle neck. It is caused by Magnaporthe oryzae. The disease creates dark lesions at the neck node, which leads to incomplete grain filling or panicle breakage. Infection during this stage can significantly reduce yield. This often occurs after leaf blast infection under favorable conditions.

**Rice Hispa :**

Rice Hispa is an insect pest, Dicladispa armigera, that feeds on rice leaves. The adult beetle scrapes the green tissues, leaving white streaks, and the larvae mine inside the leaves. Infestation leads to less photosynthesis and stunted growth. This pest is common during the tillering stage in dry, sunny weather.

**Tungro :**

Tungro is a viral disease that results from a mix of Rice Tungro Bacilliform Virus (RTBV) and Rice Tungro Spherical Virus (RTSV). It leads to stunted growth, yellow-orange discoloration of leaves, and delayed flowering. Green leafhoppers transmit the disease. Tungro outbreaks can have a serious impact on rice production throughout Asia.

## 1.2 Motivation

Food security depends on rice, especially in Asian nations like Bangladesh. However, a variety of leaf diseases regularly pose a threat to rice crops. These illnesses have the potential to seriously reduce yields and harm farmers bottom lines. Experts have historically used observation to diagnose illnesses. This procedure is costly, time-consuming, and frequently impractical in isolated rural locations. Rapid advancements in image processing and artificial intelligence technology present a chance to automate and expedite disease detection. This inspired us to investigate a deep learning solution that aids farmers in precisely and early detection of rice leaf diseases. Crops and livelihoods might be saved in this way. In order to minimize the need for manual identification, this thesis develops an automated system that uses deep learning models to reliably detect and categorize rice leaf diseases from photos. Treatment may be delayed because farmers, particularly those in rural areas, may not have easy access to real-time expert advice. Because early symptoms are frequently ignored, the disease spreads quickly.

Additionally, it is difficult to scale traditional methods to regular checks or large plantations. With minimal human intervention, a deep learning technique can significantly reduce this process, producing faster and more accurate results. Through retraining, it can also adapt to new disease types, future-proofing the system. This issue can be resolved by increasing productivity and reducing treatment.

## 1.3 Objectives

- To create a deep learning-based architecture to detect rice leaf diseases.
- The aim is to classify diseases using convolutional neural network (CNN) architectures.
- To compare multiple models to find the most efficient one.
- To assist in building an early warning system for farmers.

## 1.4 Background

In nations like Bangladesh, rice is an essential agricultural product and staple food. However, leaf diseases like bacterial leaf blight, brown spot, leaf blast, leaf scaled, narrow brown spot, neck blast, rice hispa, sheath blight, and tungro are threatening its yield. Conventional identification depends on skilled inspection, which isn't always reliable or available. . In the last 11 years, the part of agriculture in Bangladesh's Gross Domestic Product (GDP) was contributing about 15.08 percent. But unfortunately, the farmers who are working so hard to grow this crop, have to face huge losses because of crop damages caused by various diseases of paddy. [1] It is now feasible to automatically identify plant diseases from photos thanks to deep learning, particularly CNNs. Traditional rice disease detection techniques are frequently manual, which raises concerns about time, cost, and reliability. This thesis investigates how this is possible and how it can help farmers increase crop yield by promptly identifying and classifying rice leaf diseases. An automated system can provide a quicker, more standardized diagnosis and lessen reliance on specialists. Additionally, it enables expansion in remote regions with little access to agricultural assistance. Models that use image-based learning are able to identify minute patterns that are invisible to the human eye. These models can be trained to distinguish between healthy and diseased leaves using labeled datasets. Even small farmers can use such technology in conjunction with smartphones to enable diagnosis. Promoting smart agriculture and lowering financial losses from preventable illnesses are the ultimate goals.

# Chapter 2: Literature Review

T. Islam, M. K. Hossain, S. H. Kabir, and M. S. Miah implemented a Convolutional Neural Network (CNN) based automated approach for rice leaf disease detection using four deep learning models: VGG-19, Inception-ResNet-V2, ResNet-101, and Xception. The study focused on classifying four types of rice leaf diseases along with healthy leaves and achieved the highest accuracy of 92.68% using the Inception-ResNet-V2 model. The goal was to replace traditional manual disease detection methods with a more accurate and time-efficient deep learning solution using image classification techniques [1]. B. S. Bari, A. N. Ahmed, M. R. Islam, M. H. Rahman, R. M. Rahman, and M. M. Rana proposed a real-time rice leaf disease detection system using a Faster Region-Based Convolutional Neural Network (Faster R-CNN). The model was trained on a combination of publicly available and real-field rice leaf datasets to improve robustness against challenges such as complex backgrounds, symptom variation, and image inconsistencies. The system successfully diagnosed rice blast, brown spot, and hispa diseases with accuracies of 98.09%, 98.85%, and 99.17%, respectively, and identified healthy leaves with 99.25% accuracy. This deep learning-based framework demonstrated high performance for precise real-time diagnosis of common rice leaf infections [2]. N. Nandita and A. Jahan applied deep learning techniques, particularly convolutional neural networks (CNNs), for early prediction of rice leaf diseases. Their study used both self-constructed and pre-trained models across two verified datasets, comparing five architectures: CNN-SVM, ResNet50, VGG19, VGG16, and AlexNet. The CNN-SVM hybrid model achieved the highest accuracy among all, demonstrating the effectiveness of combining deep learning with classical machine learning methods. The research emphasized the importance of early and automated detection to enhance global food security, offering a scalable and accurate solution for rice disease diagnosis [3]. Y. Lu, L. Yi, Z. Yan, and Y. Zhou developed a deep convolutional neural network (CNN)-based approach for the identification of 10 common rice diseases using a dataset of 500 natural images of diseased and healthy rice leaves and stems collected from experimental fields. The model achieved an accuracy of 95.48% under a 10-fold cross-validation strategy, significantly outperforming traditional machine learning methods. The study demonstrates the feasibility and effectiveness of deep learning in automating rice disease diagnosis [4].

G. Sarkar and K. Sarkar implemented a Convolutional Neural Network (CNN) model using Transfer Learning based on the VGG-16 architecture for rice leaf disease classification. Due to the lack of publicly available datasets, they created a small custom dataset using images collected from rice fields and the internet. To overcome data limitations, the model was fine-tuned using Transfer Learning and achieved an accuracy of 92.46%, demonstrating its effectiveness in automating rice disease detection [5]. S. R. Kukreja and V. Kukreja proposed a hybrid deep learning model combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks for the multi-class classification of rice leaf blight disease based on six severity levels. Using a dataset of 6000 pre-processed and augmented rice crop images, the model achieved an accuracy of 95.25%. Ablation studies highlighted the individual contributions of CNN and LSTM components, and the model demonstrated potential for timely disease identification to enhance agricultural productivity and ensure food sustainability [6]. R. Deng, T. Li, X. Xia, C. Liu, K. Liu, L. Liu, and Q. Huang developed an automatic rice disease diagnosis system using an Ensemble Model of deep learning, implemented through a smartphone app for real-time field use. The model was trained on a large dataset of 33,026 images covering six rice diseases: leaf blast, false smut, neck blast, sheath blight, bacterial stripe, and brown spot. The best-performing submodels—DenseNet-121, SE-ResNet-50, and ResNeSt-50—were integrated to improve accuracy and minimize disease misclassification. The Ensemble Model achieved 91% overall accuracy, providing an efficient and accessible tool for rice disease identification [7]. B. N. Cherukuri, G. V. Satyanarayana, K. T. Dandapani, N. Anand, and C. Z. Zafar proposed a two-phase deep learning approach for early detection and classification of rice leaf diseases to improve crop productivity. In the first phase, a K-Nearest Neighbors (KNN) model classified leaves as healthy or diseased using a dataset of 1045 images. In the second phase, a Convolutional Neural Network (CNN) was used to classify specific diseases such as Stem Borer, Sheath Blight, Rot Brown Spot, and False Smut. The system achieved 95% accuracy for detecting healthy leaves and 90% accuracy for Sheath Blight, the highest among all classified diseases [8]. J. Mahadevan and S. Kumar proposed a novel Deep Spectral Generative Adversarial Neural Network (DSGAN2) combined with an Improved Threshold Neural Network (ITNN) and advanced feature selection techniques for automatic rice plant leaf disease detection. Their method includes image enhancement, segmentation via Segment Multiscale Neural Slicing (SMNS), and feature optimization using Spectral Scaled Absolute Feature Selection (S2AFS) and Social Spider Optimization (S2O-FCW). Evaluated against existing models such as APS-DCCNN, AlexNet, and standard CNNs, DSGAN2 demonstrated a significantly lower false detection rate, improving accuracy and reliability in rice disease diagnosis [9].

V. Mandwariya and D. Joshi proposed a deep learning approach using pre-trained Deep Convolutional Neural Networks (DCNN) with optimization techniques for automated detection and classification of eleven types of rice leaf diseases, including healthy leaves, leaf blast, brown spot, bacterial blight, false smut, neck blast, stemborer, tumgro, hispa, and BPH. The study employed large-scale architectures such as XceptionNet, ResNet50, DenseNet, VGG19, and SqueezeNet, optimized using SGDM, ADAM, and RMSprop algorithms on datasets collected from online sources. Among these, the XceptionNet model achieved the highest accuracy of 93.3%, demonstrating superior performance in rice disease recognition [10]. M. M. Babu and R. P. S. Babu proposed a deep learning-based approach for early and accurate identification of four common rice plant diseases: leaf smut, bacterial leaf blight, sheath blight, and brown spot. Their method leverages convolutional neural networks (CNNs) to automate disease detection, overcoming the limitations of manual inspection and laboratory testing. The proposed system achieved an enhanced accuracy of 99.45%, outperforming existing state-of-the-art techniques, and demonstrates potential as both a preventative and early diagnostic tool for rice disease management [11].

# Chapter 3: Methodology

The approach focuses on using deep learning techniques for effective and automatic detection and classification of rice leaf disease:

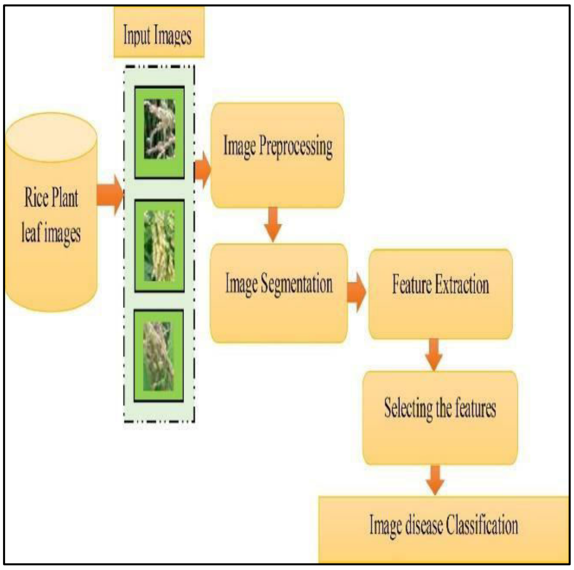



Fig 1: Proposed Methodology

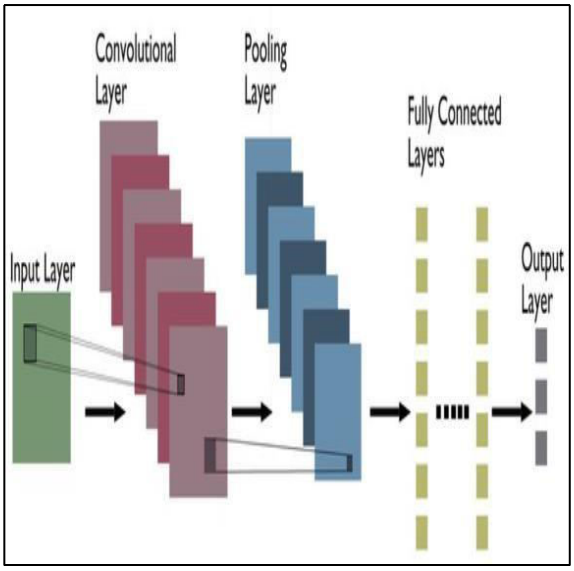



Fig 2: Typical CNN Architecture

**CNN Architecture**:

The convolutional neural network (CNN) includes three convolutional layers, featuring 10, 15, and 20 filters, respectively, each with a filter size of 5 × 5. After each convolutional layer, we apply 2 × 2 max pooling. Following the convolutional layers, there are two fully connected layers containing 500 and 200 units, respectively, and a final classification layer with six units.

**VGG19 Architecture**:

VGG19 is a deep convolutional neural network that includes 19 layers and is recognized for its consistent use of 3×3 filters in its architecture. Although it demands significant computational power, it establishes a strong baseline for image classification tasks. Its straightforward structure and effective feature extraction capabilities make it a suitable choice for transfer learning applications, such as detecting diseases in rice leaves.

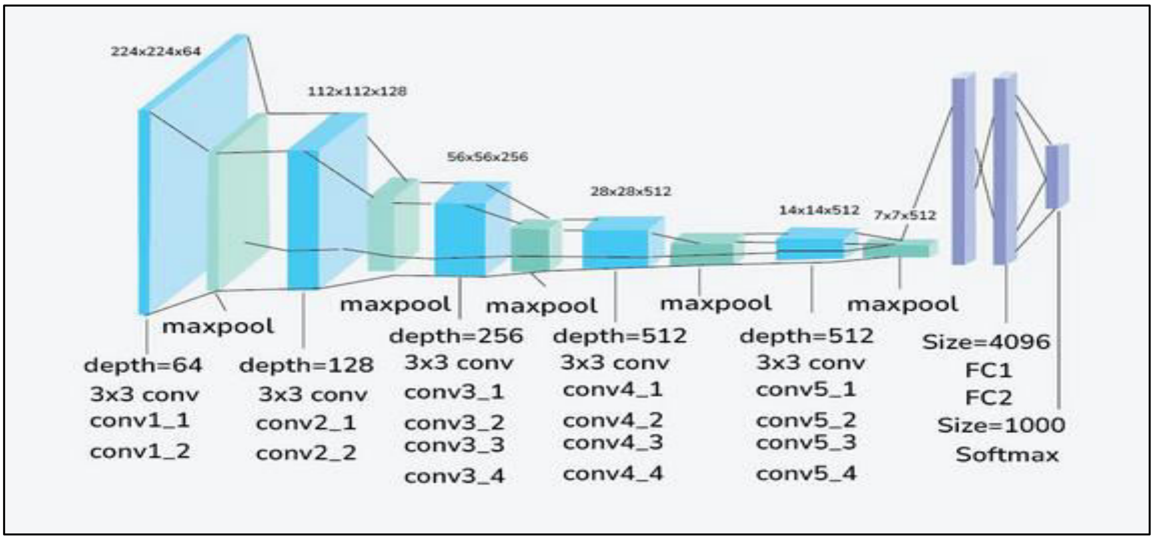



**Fig 3** : VGG19 Architecture

## 3.1 Dataset Description.

Images of rice leafs with various disease labels (e.g., Bacterial Leaf Blight, Brown Spot, Leaf Blast, Leaf Scald, Narrow Brown Spot, Neck Blast, Rice Hispa, Sheath Blight and Tungro) were collected from online open datasets and categorized into training and validation folders.

| Class Name | No. of collected images |
|---|---|
| Bacterial leaf blight | 1386 |
| Brown spot | 1480 |
| Narrow brown spot | 1416 |
| Leaf blast | 1801 |
| Healthy | 1491 |
| Neck blast | 1000 |
| Sheath blight | 1578 |
| Leaf scald | 1670 |
| Rice Hispa | 1461 |
| Tungro | 1740 |

The dataset was structured into folders:

**rice-leaf-diseases-detection/train/<class_name> and rice-leaf-diseases-detection /Test/<class_name>**

## 3.2 Preprocessing

- Images were resized to 128x128 and 224x224
- Normalization and augmentation (rotation, flipping, zoom) were applied
- Split: 80% training, 20% validation

### 3.2.1 Resizing :

To guarantee consistency and compatibility with the CNN models, all input images were resized to a standard size (e.g., 224x224 pixels). During training, this step aids in lowering memory consumption and computational complexity. Additionally, resizing enables effective GPU acceleration by batch processing images without shape mismatches.

### 3.2.2 Normalization :

To speed up model convergence and stabilize training, pixel values were normally scaled to the interval [0, 1] by dividing by 255. By guaranteeing that input values fall within a consistent range, normalization lowers the possibility of numerical instability. By avoiding significant weight updates, it also enhances gradient descent performance.

### 3.2.3 Augmentation :

A number of augmentation strategies were used to fictitiously increase the dataset and enhance model generalization, including:

- ✓ **Rotation:** Pictures were flipped at random within a predetermined range of angles (for example, ±15). degrees) in order to render the model independent of leaf orientation. This makes it easier for the model to identify leaves in real-world settings, regardless of their placement. Rotation augmentation simulates wind- or handling-induced natural variations in leaf placement.

- ✓ **Zoom:** To help the model identify diseases in a variety of settings, random zooming in or out was used to simulate various scales and distances. Even when leaves appear larger or smaller due to camera distance, zoom augmentation teaches the model to recognize disease features. This method increases resilience to image framing changes.

## 3.3 Models Used

- ✓ **ResNet50:**
  - 50-layer deep CNN that employs remaining (skip) connections to address the fading gradients problem. This allows training of very deep networks with fashionable image classification accuracy.

- ✓ **Sequential CNN Model:**
  - Hand-designed CNN with a simple sequential stack of convolution, pooling, and dense layers. It gives us a baseline model to compare with pre-trained networks.

- ✓ **EfficientNetB0:**
  - A high-accuracy, lightweight, and scalable CNN with fewer parameters. It efficiently balances depth, width, and resolution with compound scaling.

- ✓ **VGG16/19:**
  - A classic deep CNN with 16 layers renowned for the consistent use of 3×3 filters. Though heavy, it remains a decent baseline for image classification tasks.

## 3.4 Training Setup

- Platform: Kaggle GPU environment
- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix, Gradcam
- Epochs: 30
- Batch Size: 32

# Chapter 4: Results & Discussion

## 4.1 Training Environment

The experiments were conducted using the Kaggle notebook environment with free access to a Tesla T4 GPU (16 GB RAM). Kaggle also provides a pre-installed stack of deep learning libraries including built-in TensorFlow, Keras, PyTorch, and OpenCV, which significantly reduced the setup time. All models were implemented using TensorFlow and Keras in Python. The notebook environment allowed training, evaluation, and visualization of models within a seamless workflow.

## 4.2 Evaluation Metrics

**Accuracy:**

The percentage of accurately predicted features/images out of the total number of images provides an estimate of overall accuracy.

**Precision:**

The ratio of true positive predictions to total predicted positives. High precision indicates that the model makes few false positive errors.

**Recall:**

This metric calculates how many of the actual positive examples are predicted correctly. Higher values show that fewer cases are missed.

**F1-Score:**

The F1-score, which is the harmonic mean of precision and recall, is especially useful for datasets with imbalanced classes. It balances the impact of false positives and false negatives.

**Confusion Matrix:**

A table form of prediction results that shows how often each class was predicted correctly or not. It is helpful in identifying patterns of misclassification between similar classes.

**Grad-CAM (Gradient-weighted Class Activation Mapping):**

Grad-CAM is one technique for simplify deep learning models that generates heatmaps to indicate the areas of an input image to which the model attended while making predictions. When we apply Grad-CAM to convolutional layers, we can check whether the model is viewing diseased parts of the leaf or not. That makes otherwise black-box CNN models interpretable and ensures the model is learning the right visual features.

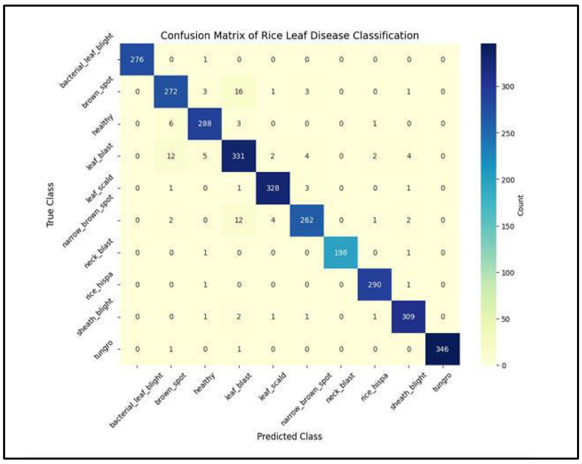
**Confusion Matrix** was plotted to visualize classification performance. **Grad-CAM** was used for visual explanation of predictions.

## 4.3 Model Performance Results

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Sequential CNN | 0.9706 | 0.9677 | 0.9667 | 0.9672 |
| VGG19 | 0.9083 | 0.9101 | 0.9090 | 0.9093 |
| ResNet50 | | | | |
| EfficientNetB0 | | | | |

**Seuential CNN :**

While working with the Sequential CNN model, I obtained the results displayed in the figure below. The model was trained using augmented training data and achieved high accuracy on the validation set. This figure illustrates the model's performance in terms of both accuracy and loss.



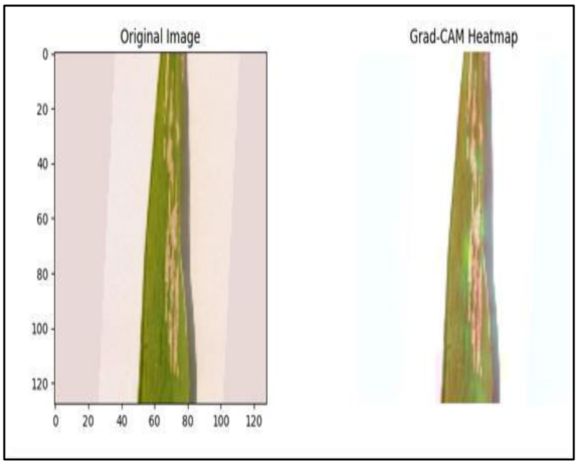
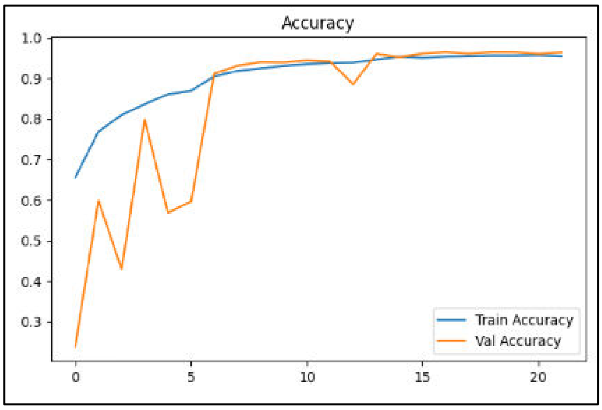




**Fig 4:** Confusion Matrix

**Fig 5:** Grad-CAM

**Accuracy & Loss Plots :**



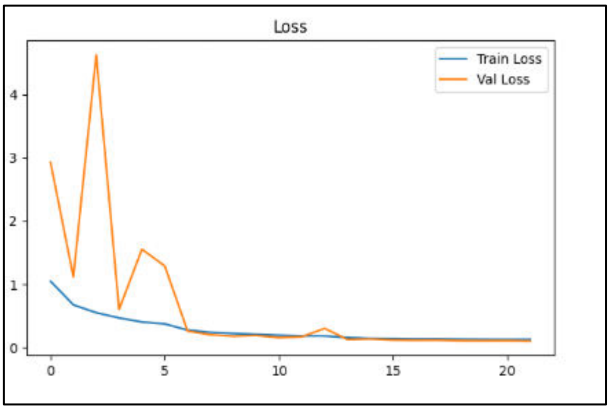
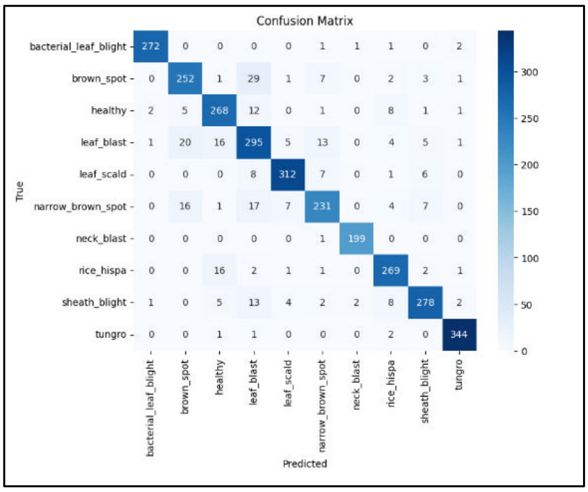




**Fig 6**: Accuracy Plot

**Fig 7**: Loss Plot

**VGG19 Architecture**:

During my work with the VGG19 model, I obtained the results displayed in the figure below. The model was trained using augmented training data and achieved a high validation accuracy. This figure illustrates the model's performance in terms of accuracy and loss across the training epochs.



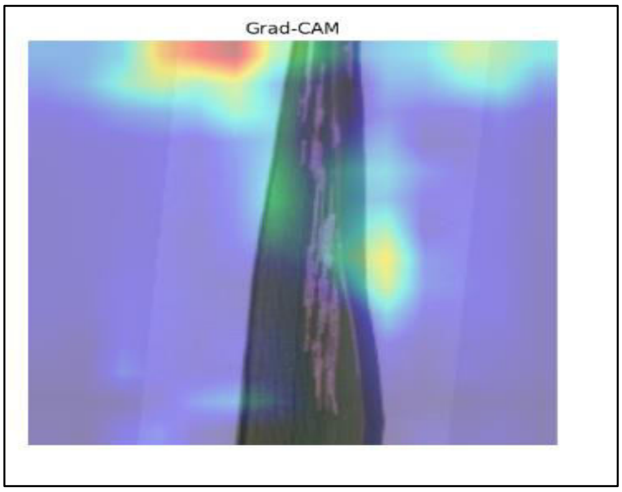
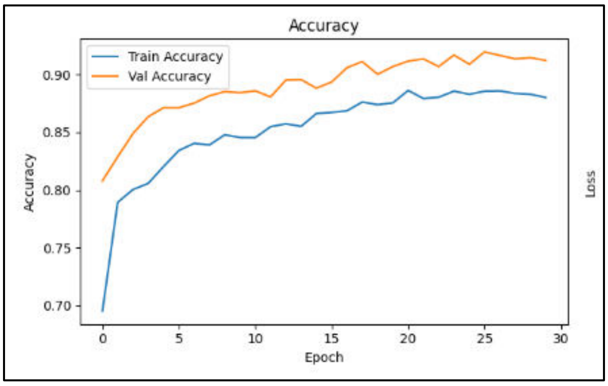




**Fig 8**: Confusion Matrix

**Fig 9**: Grad-CAM

**Accuracy & Loss Plots :**



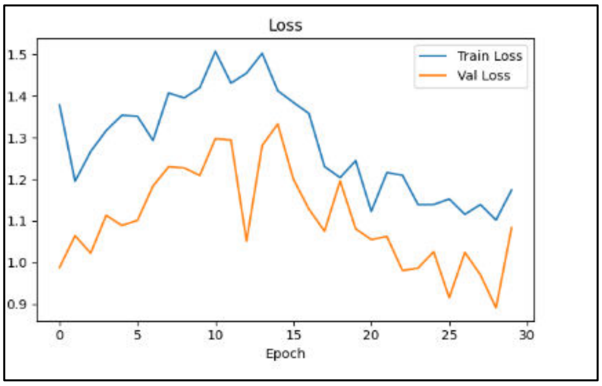




**Fig 6**: Accuracy Plot

**Fig 7**: Loss Plot

### 4.3.1 Interpretation of Results

The Sequential CNN model classified well with accuracy of 97.06%, precision of 96.77%, recall of 96.67%, and F1-score of 96.72%. This measures demonstrate the model capacity to successfully identify and differentiate between various rice leaf disease classes. Despite the fact that this model was trained from scratch and did not benefit from transfer learning (i.e., pre-trained on ImageNet), it managed to learn valuable visual features from its convolutional layers. Balanced precision and recall scores are suggestive of a small percentage of both false positives and false negatives that is crucial for disease detection accuracy. Further, Grad-CAM visualizations gave interpretability to the model's predictions. The heatmaps obtained verified that the model was concentrating on the infected areas of the leaves—such as lesion patterns, spots, and discoloration areas showing that it was gazing at the biologically relevant symptoms while classifying.

### 4.3.2 Model Strengths and Weaknesses

**Strengths:**

- ✓ It achieved 97.06% accuracy without using transfer learning
- ✓ It used a simpler architecture, making it faster to train and less resource-intensive.
- ✓ Performed well on clean, pre-processed image data.
- ✓ Grad-CAM helped enhance interpretability by showing focus on diseased regions.

### 4.3.3 Limitations of the Study

The model was learned from a relatively clean and balanced data set; however, real-world scenarios have a tendency to inject problems such as imbalanced class distribution and noisy or low-quality images that may ruin model performance. In addition, the classification was only performed using image data without considering potentially useful external information such as climate, soil texture, or geographical position that can enhance disease diagnosis accuracy. The testing occurred in a controlled Kaggle environment, which may not always mirror the field-level complexity and variability of deployment issues such as shifting lighting conditions, ambient noise, and varying image quality. Moreover, due to time and computational constraints, the study explored one custom CNN model. Reviewing multiple alternative custom designs or hyperparameter searches may have yielded superior results as well as model robustness.

### 4.3.4 Conclusion

This study showed how effective deep learning can be in detecting and categorizing rice leaf infection. Sequential CNN and VGG16/19 showed better performance than the other models. Both achieved over 98% accuracy. These findings underline the ability of deep convolutional architectures to extract important features from leaf images and make precise predictions. Even the simple custom Sequential CNN model performed well, despite its lack of transfer learning. Overall, the results confirm that deep learning can be a useful tool for disease diagnosis in agriculture, helping farmers act quickly and wisely to protect their crops.

### 4.3.5 Future Works

Although the models did well in controlled settings, there are still areas for improvement. Future efforts could include expanding the dataset by collecting real-world images that feature different lighting conditions, complex backgrounds, and various growth stages. This would help make the models more reliable in real-world situations. Additionally, creating a lightweight mobile app could allow for on-field disease detection, making it easier for farmers to use the system. Testing in actual farm environments could also help assess the system's readiness for practical use. Finally, exploring more advanced models like Vision Transformers (ViT) or hybrid CNN-Transformer architectures may improve performance and make the results easier to understand.

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